

The empirical recurrence for OEIS A300884, recovered from its tail

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Abstract

OEIS A300884 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 118$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 101$. Its last coefficient is $c_{96} = -4 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300884 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 3, 4 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 3, 8, 40, 121, 465, 1822, 7045, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Mar 14 2018 07:26:53, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 118$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 236$ exact terms from $n = 5$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	1	c_{25}	104873	c_{49}	74094982	c_{73}	−993006
c_2	10	c_{26}	2432903	c_{50}	36958660	c_{74}	−101400
c_3	16	c_{27}	3733046	c_{51}	−20390098	c_{75}	501922
c_4	−9	c_{28}	1659039	c_{52}	−56838798	c_{76}	20044
c_5	−123	c_{29}	−4497143	c_{53}	−50189681	c_{77}	−170463
c_6	−133	c_{30}	−11037080	c_{54}	−10262611	c_{78}	75432
c_7	221	c_{31}	−11531701	c_{55}	31206563	c_{79}	98512
c_8	723	c_{32}	−2358495	c_{56}	41472529	c_{80}	−30932
c_9	667	c_{33}	11602884	c_{57}	16283553	c_{81}	−15587
c_{10}	−1522	c_{34}	22291397	c_{58}	−14046930	c_{82}	27659
c_{11}	−4439	c_{35}	16968730	c_{59}	−27332609	c_{83}	1518
c_{12}	−4077	c_{36}	−4518486	c_{60}	−17414838	c_{84}	−16513
c_{13}	3995	c_{37}	−29287578	c_{61}	4240780	c_{85}	634
c_{14}	21125	c_{38}	−38300589	c_{62}	20534324	c_{86}	7405
c_{15}	22115	c_{39}	−16959205	c_{63}	15386380	c_{87}	−424
c_{16}	−24164	c_{40}	20111625	c_{64}	−4305682	c_{88}	−2604
c_{17}	−88412	c_{41}	55548085	c_{65}	−12917416	c_{89}	−8
c_{18}	−72717	c_{42}	55255377	c_{66}	−3023924	c_{90}	936
c_{19}	78380	c_{43}	14789432	c_{67}	4903097	c_{91}	21
c_{20}	327020	c_{44}	−45700866	c_{68}	1688816	c_{92}	−260
c_{21}	410383	c_{45}	−73185335	c_{69}	−2049914	c_{93}	−60
c_{22}	−78957	c_{46}	−53218996	c_{70}	−635398	c_{94}	32
c_{23}	−855941	c_{47}	4758018	c_{71}	979566	c_{95}	16
c_{24}	−1094570	c_{48}	57862530	c_{72}	−158516	c_{96}	−4

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 101$, and it is the recurrence OEIS A300884 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{96-1} - \dots - c_{96}$. Its constant term is $-c_{96} = 4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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